Lesson 4: Direct Hit

001 Game Creator (Windows only)

VIDEO GAME CREATION

This week: detecting when things touch. Your job is to add one enemy to the screen and make your bullet destroy it on contact - the first piece of your capstone's actual combat loop.

Open your own [YourName]Game project from Lesson 3 - don't start a new one. Everything this week builds on the player and bullet you already have.

Tip: this lesson's Assets folder has a ready-to-use placeholder enemy image (enemy_invader_red.png) and a new sound effect (sfx_hit.wav) - you don't need to find or make your own art today.

Setup

1. Open your [YourName]Game project from Lesson 3.
2. Create a new Actor Template for the enemy (the Character template works well) using enemy_invader_red.png (import it from this lesson's Assets folder) as its Body sprite, and name it enemy.
3. Place one enemy Actor on the Map near the top of the play area - leave "Initially Existent" ticked this time, since (unlike the bullet) there's only ever one enemy and it should be there from the start.

Level 1 - Minimal (MVP)

A single enemy sits on screen. When your bullet touches it, both the bullet and the enemy disappear. This is a complete finish for today - reaching this is enough to move on to next week.

4. On the enemy Actor, add an Actor Trigger: "Collided with actor" -> a script with two "Actor – Destroy/Revive Actor" events (Destroy): one targeting relate (the actor that collided with it - your bullet) and one targeting this (itself).
5. Save, then Play/Test and shoot the enemy - both the bullet and the enemy should disappear on contact.

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enemy Actor Trigger: Collided with actor ->
  Actor – Destroy/Revive Actor(relate, Destroy)
  Actor – Destroy/Revive Actor(this, Destroy)
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Alternate Path: Textual Scripting (typed BASIC-like code)

The same collision-detection pattern works with typed commands instead of graphical event blocks, in a separate throwaway project (see Lesson 3's warning about this switch being permanent).

IMPORTANT: only try this in your separate throwaway test project from Lesson 3, never your main [YourName]Game project.

6. Open your throwaway Textual-mode project from Lesson 3 (or create a new one, following Lesson 3's Setup steps).
7. Open the Script Editor for the enemy Actor's "Collided with actor" trigger, same as before.
8. Instead of dragging graphical event blocks, type the command directly - start typing "Destroy" and use the editor's autocomplete/command list to find the text equivalent of the Actor – Destroy/Revive Actor event you used in the graphical path, once for relate and once for this.

The Level 3 sound idea maps onto Textual commands too - it's a typed equivalent of Play Sound. Exact command names should be checked against the installed version's Script Editor autocomplete or the wiki before class.

Level 2 - Complete (Alpha/Beta)

The enemy is destroyed cleanly and exactly once, even if you fire several bullets in a row - no errors, no lingering references, and a bullet that misses still behaves exactly like it did last week.

- Fire several bullets in a row, including some that miss - each hit destroys exactly one bullet and the enemy, and any bullet that misses keeps flying and destroys itself off-screen exactly like it did in Lesson 3.
- Once the enemy Actor is destroyed, it no longer exists to collide with anything, so its own "Collided with actor" trigger can't run a second time - the enemy can only ever be destroyed once, even if two bullets somehow reach it on the same update.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Sound: import sfx_hit.wav (Resources > Import > Sound), then add a "Play Sound" event to the enemy's "Collided with actor" script, before the two Destroy events.
- Visual: give the enemy Actor its own Tint/Opacity setting (Actor > Graphics Options), distinct from the bullet's, so it clearly reads as a threat rather than another shot.
- Structure: keep everything about how the enemy responds to being hit - the sound, both Destroy events - on the enemy Actor's own "Collided with actor" trigger. The bullet Actor doesn't need any new scripting at all this week; it only needed to exist as something the enemy's trigger could detect.

If Something's Wrong

- Nothing happens on contact: confirm the trigger is "Collided with actor" (not "Collided with solid", which is for walls/terrain), and that both the enemy and bullet Actors have "Solid" ticked - non-Solid Actors generate no collisions at all.
- Error or nothing happens when destroying relate: confirm relate is actually available in this trigger's script - it refers to whichever Actor collided with the one running the script, and only has a meaningful value inside triggers like this one.
- Enemy destroys itself immediately, with no bullet involved: check that the enemy and bullet Actors aren't overlapping in their initial placement on the Map - move the enemy's starting position if needed.

Want More?

001gamecreator.com/wiki, Student Resource List > 001 Game Creator